Balanced districting for a national sales channel: the model, the two-stage scheme, and the balance ceiling

Territory-division project*

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Abstract

The territory problem has been re-scoped. Instead of dividing a contested region between two legacy representatives of merging firms, the business is standing up a *new* "national" channel — the two largest product manufacturers carved out of the financial-institutions and wirehouse channels — and wants it cut into territories of roughly equal market opportunity. This note develops the mathematics of that problem. We generalise the utility model to n representatives per postal code and show it reduces identically to the two-player model at $n = 2$; we prove the model is invariant under a global rescaling of the dollar fields, which licenses exporting a descaled instance from the confidential machine; we prove the central structural fact, that *maximum Nash welfare on a common measure is exactly equal-size districting*, so the ∼\$1B target is an emergent property rather than a constraint; we decompose total welfare into a partition-invariant part and an incumbency premium, and size the two on real saturation; and we prove that a *disconnected* footprint — the channel covers the two coasts, Texas and Florida but not the midwest — imposes a computable *balance ceiling* that bounds every partition from above, is available before any solver runs, and already shows that a ±10% band around \$1B is probably geometrically unreachable. We close with the stage-1 mixed-integer formulation and the open modelling questions.

1 The instance and the re-scoping

The programme's original problem was bilateral. Two sales forces merge; a national *census* intersects the two legacy representative maps and decomposes the country into contested components, each with one A-rep and one B-rep; inside a component the zips are divided between them by maximum Nash welfare subject to hard contiguity. Everything downstream — one binary per zip, a scalar exchange rate $u_a(z)/u_b(z)$, the separator cut with its "a branch" and its "b branch" — inherits that two-sidedness.

The new problem is not a merger. A *national channel* is being created by carving the two largest manufacturers out of the existing financial-institutions and wirehouse channels, and its footprint is to be divided into territories of roughly equal opportunity. Two structural consequences follow. First, *there is no pair structure to decompose along*: the census's components

*A self-contained companion to `nash_territory_division.tex` and to `research/contiguity/math_note/math_note.tex`, which treats the two-player merger problem this note re-scopes. The source of record is `research/contiguity/NWAY.md`; the implementations are `td/model.py` and `td/channel.py`, with `battery/code/tests/test_nway.py` and `test_channel.py` asserting the identities proved below. Every number in Section 7 and in the saturation arithmetic of Section 5 is computed by `ceiling.py` and injected as a macro, never transcribed.

came from the overlap of two legacy maps, and a greenfield channel has no such overlap, so what was many independent 30–500 zip subproblems becomes one k-way partition of the whole footprint. Second, *the objective's argument changes measure*: what is to be balanced is *opportunity*, a quantity attached to the territory and not to any representative — the same for every candidate owner. That single change turns the Nash criterion into a balance criterion exactly (Section 4), which is the main result of this note.

zips carrying sales	2,232
distinct representative keys	72, including an “open” vacancy key $\Rightarrow$ 71 real
total market opportunity	$\approx$\$6.2B
footprint	west coast, east coast, Texas, Florida; **midwest uncovered**
territory target	$\sim$\$1B $\Rightarrow k \approx 6$ (or 7 at $\sim$\$886M)

Three readings matter. First, *scale sits at the frontier but not past it*: 2,232 units against 6 districts is $\approx$13,392 assignment binaries, the same order of magnitude as the $\sim$1,500 units Validi and Buchanan (2022) certify for political districting with lazily separated connectivity cuts, and an order of magnitude above the 135 zips the programme's own single-tree solver currently certifies. Attemptable, not routine.

Second — and this is the useful one — *the footprint is disconnected.* No contiguous territory spans California and Florida, so the adjacency graph has at least 4 major components and the problem *separates*: give each component an integer district count, then solve each independently at $\sim$500 zips rather than 2,232 at once. Failure mechanism (a) of the two-player programme — pre-existing graph disconnection, the thing that made the legacy cut loop's dual bound unsound — arrives here as the structure that makes the problem tractable. It also imposes a hard limit on achievable balance, which Section 7 makes precise.

Third, 71 representatives against $\approx$6 territories is a 12:1 ratio. The reading that makes sense of it is that the channel is a *specialist carve-out* staffed by a handful of senior wholesalers while the remaining representatives keep covering the other manufacturers in the existing channel; the unmatched representatives of Section 6 are therefore *not selected for this channel*, not released. If that reading is wrong, stage 2's framing needs revisiting.

2 The utility model with n representatives

2.1 Definition

Fix a zip z. Let $S_i(z) \geq 0$ be representative i's booked production at z, let

$$T_z \;=\; \sum_j S_j(z) \tag{1}$$

be all production booked to a named representative there, let $S_{\text{free}}(z) \geq 0$ be production booked at z under no named representative (Section 2.3), and let $M_z > 0$ be the market opportunity. Two parameters translate these into value: the *transfer capture* $\theta \in [0, 1]$, the fraction of another representative's book an inheriting representative retains, and the *headroom credit* $\lambda \in [0, 1]$, the rate at which untapped opportunity counts against booked production. With the net headroom convention $c_1 = 1 - \lambda$, $c_2 = \theta(1 - \lambda)$,

$$\boxed{u_i(z) \;=\; c_1\, S_i(z) \;+\; c_2\big(T_z - S_i(z)\big) \;+\; c_{\text{free}}\, S_{\text{free}}(z) \;+\; \lambda M_z} \tag{2}$$

The inheriting representative keeps c_1 of their own book, captures c_2 of everyone else's, capitalises unowned book at a rate c_{free} decided separately, and is credited λ of the whole opportunity. Note that T_z in (1) runs over named representatives only; S_{free} enters (2) through its own term.[1]

Headroom generalises verbatim: the model requires

$$M_z \;\geq\; \max_i \big(S_i(z) + \theta\,(T_z - S_i(z))\big). \tag{3}$$

An *allocation* is a map own : $Z \to R$ from zips to representatives, and each representative's *gain* at disagreement point $d = 0$ is the bundle utility

$$g_i \;=\; \sum_{z:\,\text{own}(z)=i} u_i(z). \tag{4}$$

The criterion is maximum Nash welfare (MNW), the Nash bargaining solution (Nash, 1950) at $d = 0$, equivalently the Eisenberg–Gale objective (Eisenberg and Gale, 1959):

$$\max_{\text{own}} \sum_i \log g_i \qquad \text{subject to per-representative contiguity.} \tag{5}$$

Nothing in the outer-approximation argument that makes (5) exactly solvable was ever two-player: log is concave, g_i is linear in the assignment, so this remains a convex MINLP with a genuine optimality certificate (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994). The fairness guarantee survives the move for the same reason — Caragiannis et al. (2019) state Pareto efficiency and envy-freeness up to one good for n agents, not two.

2.2 The model reduces to the two-player one

Proposition 1 (Two-representative reduction)**.** *Let $n = 2$ with $R = \{a, b\}$, $S_a(z) = A_z$, $S_b(z) = B_z$ and $S_{\text{free}} \equiv 0$. Then (2) is identically the two-player model,*

$$u_a(z) = c_1 A_z + c_2 B_z + \lambda M_z, \qquad u_b(z) = c_2 A_z + c_1 B_z + \lambda M_z,$$

and (3) is identically $M_z \geq \max(A_z + \theta B_z,\ B_z + \theta A_z)$.

Proof. $T_z = A_z + B_z$, so $T_z - S_a(z) = B_z$ and $T_z - S_b(z) = A_z$; substituting into (2) with $c_{\text{free}} S_{\text{free}} = 0$ gives the two displayed expressions. For (3), the maximum runs over $i \in \{a, b\}$ and gives $\max(A_z + \theta B_z,\ B_z + \theta A_z)$. □

This is not a formality. It is what keeps the entire committed corpus of two-player results — the paper's worked 50-zip instance, the C1–C9 battery, every certified harness row — interpretable under the new model, and it is asserted numerically rather than argued: `test_nway.py::test_two_rep_reduction` requires every n-way primitive to agree with the frozen two-player `base.py` to floating-point equality on a two-representative instance.

[1] The implementation follows this exactly (`nway.utilities`). Its headroom validator `nway.headroom_violations` is deliberately *more* conservative: it folds S_{free} into T_z and treats the filler as a possible holder, so a zip whose orphaned book alone exceeds its opportunity is still flagged.

2.3 Unowned book: the vacancy filler

Some territories currently have no assigned representative, and their production is booked under a *filler key*: a representative-shaped sentinel that carries real sales, real opportunity and a real manufacturer, but is not a person. It must never become a candidate owner. A filler key with an objective term would have the solver bargaining on behalf of an empty chair, and because $\sum_i \log g_i$ is unbounded below as any $g_i \to 0$, it would trade real representatives' welfare away to feed it.

The schema separates the two roles, which is what makes the exclusion cheap: T_z is computed from *all* books, but the assignment ranges over *candidates* only, and candidacy is $\mathrm{cand}(z) = \{i : S_i(z) > 0\}$ with filler keys removed. Orphaned production therefore arrives as its own attribute S_{free} and is capitalised by whoever inherits the zip. Four node classes result:

class	$\lvert\mathrm{cand}(z)\rvert$	book	meaning
contested	≥ 2	real	the decision problem
uncontested	1	real	owner forced; no binary, but carries utility and adjacency
vacant	0	filler only	real book, no incumbent — nobody can claim it by legacy
untapped	0	none	opportunity with no book at all

Choosing c_{free}. Three coefficients are defensible, and they give materially different maps, so the implementation takes the mode explicitly (`filler_capture`) rather than defaulting quietly. `theta` ($c_{\mathrm{free}} = c_2$) discounts orphaned book exactly like a live representative's; conservative, and it assumes vacant business is as person-sticky as anyone else's. `opportunity` ($c_{\mathrm{free}} = \lambda$) treats it as untapped market; defensible if the sales in vacant territories are house or inbound business with no relationship content. `full` ($c_{\mathrm{free}} = c_1$) is *the recommended choice*: the reason $\theta < 1$ exists at all is that a *departing* representative pulls relationships away with them, and a vacancy has nobody left to pull — whatever book has survived an already-departed representative has, by definition, survived the departure, so discounting it again double-counts an attrition that has already happened. The current default is `theta`, only because it is the no-change case. It is probably not the right answer.

3 Scale invariance, and what it licenses

The real instance lives on a confidential machine and its dollar amounts cannot travel. The following proposition is why they do not have to.

Write the utility (2) in *share* form. With $s_i(z) = S_i(z)/M_z$, $t_z = T_z/M_z$ and $f_z = S_{\mathrm{free}}(z)/M_z$,

$$u_i(z) \;=\; M_z\Big[\,c_1 s_i(z) + c_2\big(t_z - s_i(z)\big) + c_{\mathrm{free}} f_z + \lambda\Big]. \tag{6}$$

Every dollar amount has been pushed into the single factor M_z.

Proposition 2 (Scale invariance of the model)**.** *Let $\mathcal{I} = \big(G, (S_i)_i, S_{\mathrm{free}}, M\big)$ be an instance and, for $\kappa > 0$, let $\kappa\mathcal{I}$ scale every dollar field by κ (the graph, the candidate sets and $\theta, \lambda, c_{\mathrm{free}}$ unchanged). Then for every allocation* own *and every $\rho \geq 0$:*

1. $u_i^{\kappa\mathcal{I}}(z) = \kappa\, u_i^{\mathcal{I}}(z)$ *and* $g_i^{\kappa\mathcal{I}} = \kappa\, g_i^{\mathcal{I}}$*;*
2. $\sum_i \log g_i^{\kappa\mathcal{I}} = \sum_i \log g_i^{\mathcal{I}} + n \log \kappa$*, an additive constant independent of* own*;*

3. *consequently* own *maximises* $\sum_i \log g_i - \rho\, P(\text{own})$ *on* $\kappa\mathcal{I}$ *if and only if it does on* $\mathcal{I}$*, and every objective* difference *— hence every optimality gap and every certificate width in nats — is identical on the two instances;*

4. *the headroom condition* (3) *holds on* $\mathcal{I}$ *iff it holds on* $\kappa\mathcal{I}$*.*

Proof. (1) Each of the four terms of (2) is homogeneous of degree one in the dollar fields, so u_i is; g_i is a sum of such terms over a fixed set, hence also. (2) Take logarithms: $\log(\kappa g_i) = \log g_i + \log \kappa$, and sum over the n representatives. (3) The perimeter $P(\text{own})$ is a count of boundary edges and does not involve the dollar fields at all, so the full objective shifts by the constant $n \log \kappa$; adding a constant changes neither the argmax nor any difference of objective values. (4) Both sides of (3) are homogeneous of degree one. □

This is what makes a *descaled export* exact rather than approximate. The exporter emits shares $s_i(z) \in [0, 1]$ and a relative opportunity $m_{\text{rel}}(z) = M_z/\kappa$, κ being the median positive opportunity, and refuses to write a currency amount or the divisor. The loader reconstructs $S_i = s_i \cdot m_{\text{rel}}$ and $M = m_{\text{rel}}$, i.e. exactly $\mathcal{I}/\kappa$ up to six-significant-figure rounding, so by Proposition 2 it has the *same* optimal allocations, gaps and certificates as the real instance. Removing the scale drops a constant the solver never reads. That is strictly better than the synthetic twin it supersedes: real ZCTAs, real adjacency, real territory structure, real share patterns, and no rank-jitter or fidelity-audit apparatus at all.

Remark 1 (What *is* scale-dependent)**.** Part (3) of Proposition 2 holds for every $\rho \geq 0$, including $\rho > 0$. This corrects a claim recorded in `NWAY.md` §5b, that a positive compactness weight "mixes a log-scale term with a raw perimeter count and is therefore scale-dependent". For the Nash objective it is not: the log-sum's response to a rescale is a constant, which cancels in every comparison. Three genuinely scale-dependent things remain, and they are the ones to carry forward. *(i) The other districting entry point:* `districting.solve_contiguous` minimises a Kalai–Smorodinsky gap *plus* ρ·perimeter with the gap in raw gain units (Kalai and Smorodinsky, 1975); there κ multiplies one term and not the other, so that objective's argmax really does move. *(ii)* ρ *is a calibration constant, not a model parameter:* the battery's $\rho = 2 \times 10^{-3}$ knee prices one boundary edge against a fixed number of nats on a particular family of synthetic instances, and nats per edge depends on n and on the value distribution, neither of which survives the move to this footprint. Any ρ carried over must be re-derived, as must the units of the travel-cost weight explored separately. ($\rho = 0$ is the model; contiguity is a hard constraint.) *(iii) Solver tolerances are absolute:* a feasibility tolerance of 10^{-9} is stated in gain units and gains scale with κ, whereas the certificate tolerance of 10^{-8} nats does not. Normalising M by its median keeps gains at order n rather than order 10^9, which is the numerically favourable end of that trade and an additional reason to descale.

One further caveat: the heavy-tail calibration of the synthetic generator (double Pareto-lognormal noise on sales *values*) does not transport either. Shares are bounded in $[0, 1]$ with quite different tail behaviour, so those knobs need recalibrating, not re-pointing.

4 Nash welfare on a common measure is equal-size districting

This is the centrepiece. In the two-player problem the objective and the balance of the two territories were different things; the Nash criterion was chosen for its axioms and its efficiency, and balance was a by-product. On a *common measure* they coincide exactly.

Let $G = (Z, E)$ be the footprint graph with $M_z > 0$, and let $\mathcal{A} = (A_1, \ldots, A_k)$ be a partition of Z into k districts, with masses

$$M_j \;=\; \sum_{z \in A_j} M_z.$$

Proposition 3 (MNW on a common measure equalises)**.** *For every partition of Z into k parts, $\sum_j M_j = M(Z) := \sum_{z \in Z} M_z$ — the total is* partition-invariant*. Consequently*

$$\sum_{j=1}^{k} \log M_j \;\le\; k \,\log \frac{M(Z)}{k}, \tag{7}$$

with equality if and only if $M_1 = \cdots = M_k = M(Z)/k$. Hence a partition attaining equal masses, when one exists, is a maximiser of the Nash objective, and every maximiser is as close to equal as the feasible set permits.

Proof. Partition-invariance is immediate: each z lies in exactly one part, so

$$\textstyle\sum_j M_j = \sum_j \sum_{z \in A_j} M_z = \sum_{z \in Z} M_z.$$

For the inequality, relax to $m \in \mathbb{R}^k_{>0}$ with $\sum_j m_j = M(Z)$ and maximise $\sum_j \log m_j$. The objective is strictly concave and the constraint set is compact and convex, so the maximiser is unique. The Lagrangian $L(m, \mu) = \sum_j \log m_j - \mu\big(\sum_j m_j - M(Z)\big)$ is stationary when $1/m_j = \mu$ for every j, so $m_j = 1/\mu$ for all j, and the constraint forces $m_j = M(Z)/k$. Substituting gives (7). Equivalently and without calculus, AM–GM gives $\big(\prod_j m_j\big)^{1/k} \le \frac{1}{k} \sum_j m_j = M(Z)/k$ with equality iff all m_j are equal; take logarithms and multiply by k. Since every partition's mass vector is feasible for the relaxation, the bound applies to it, and the equality case identifies exactly the equal partitions. □

So the Nash objective *is* the balance objective on a common measure. It is not an approximation of it, and it is not a proxy: the argmax is the same. That is the whole justification for treating \$1B as an emergent target rather than a hard band — set $k = \lceil \text{total opportunity}/\$1\text{B} \rceil$ and balance falls out of the criterion the project already uses for fairness.

Two refinements make the statement useful when perfect equality is unattainable, which on a real graph it always is.

Corollary 4 (Strict Schur-concavity: the objective always prefers the flatter vector)**.** $\Phi(m) = \sum_j \log m_j$ *is strictly Schur-concave on $\mathbb{R}^k_{>0}$. Hence if the mass vector m of one partition* majorises *the mass vector m' of another — m is the less even of the two, in the standard partial order — then $\Phi(m) \le \Phi(m')$, strictly unless m is a permutation of m'.*

Proof. Φ is symmetric and strictly concave, being a sum of a strictly concave function applied coordinatewise. A symmetric concave function is Schur-concave: for $i \ne j$ the Schur–Ostrowski condition $(m_i - m_j)\big(\partial_i \Phi - \partial_j \Phi\big) \;=\; (m_i - m_j)\big(1/m_i - 1/m_j\big) \;=\; -\,(m_i - m_j)^2/(m_i m_j) \;\le\; 0$ holds, with equality only at $m_i = m_j$; strictness of the inequality off the diagonal gives strict Schur-concavity. □

Corollary 4 is what makes (5) a usable objective on a constrained feasible set: among contiguous partitions the criterion orders any two comparable mass vectors by evenness, so the solver's search is a search for balance even when the perfectly balanced vector is not attainable.

Observation 1 (Verified by enumeration)**.** *`test_channel.py`'s `test_nash_welfare_is_equal_size_districting` brute-forces every way of cutting a 12-vertex path with uniform opportunity per zip into three* contiguous *blocks and checks that the argmax of $\sum_j \log M_j$ is the balanced cut $(4, 8)$, with reported relative spread exactly zero. It is a small check, but it is the one that would fail first if the identity were ever broken by a change to the objective.*

Why not just minimise a spread? Because minimising a dispersion measure is not an efficiency criterion and can leave *everybody* worse off — the project's trap 2. Re-verified at $d = 0$ on the paper's 50-zip instance: an unconstrained equalising MILP over all subsets reaches a Kalai–Smorodinsky gap of 5.2×10^{-8} at 98.1% of attainable welfare with gains (7.3226, 7.0699), and is *Pareto-dominated* by Nash's (7.3715, 7.2318) — both parties strictly prefer the Nash allocation to the "perfectly fair" one. Proposition 3 gives the balance without the pathology, because a strictly concave increasing objective stays on the Pareto frontier by construction. A hard band \$1B $\pm\, \epsilon$ would import the pathology back as an infeasibility risk, and Section 7 shows it may be unattainable anyway.

5 The welfare decomposition

Proposition 3 concerns a common measure. Real utilities (2) are not common — they differ across representatives through the book terms. The following decomposition says exactly how much that matters.

Proposition 5 (Welfare decomposition)**.** *For any allocation* own *of all of Z,*

$$\sum_i g_i \;=\; \underbrace{\sum_{z \in Z} \Big[\lambda M_z + c_2 T_z + c_{\text{free}} S_{\text{free}}(z)\Big]}_{W_0,\ \textit{partition-invariant}} \;+\; \underbrace{(c_1 - c_2) \sum_{z \in Z} S_{\text{own}(z)}(z)}_{\textit{incumbency premium}} . \tag{8}$$

The first term does not depend on own *at all. The second, since $c_1 - c_2 = (1-\theta)(1-\lambda) \geq 0$, is maximised by giving every zip to the candidate holding the most book there.*

Proof. By (4), $\sum_i g_i = \sum_i \sum_{z:\text{own}(z)=i} u_i(z) = \sum_{z \in Z} u_{\text{own}(z)}(z)$, since each z contributes to exactly one representative. Expanding (2) at $i = \text{own}(z)$ and regrouping,

$$u_{\text{own}(z)}(z) = c_2 T_z + c_{\text{free}} S_{\text{free}}(z) + \lambda M_z + (c_1 - c_2) S_{\text{own}(z)}(z),$$

in which only the last term mentions own. Summing over z gives (8). For the maximisation, $c_1 - c_2 = (1-\lambda) - \theta(1-\lambda) = (1-\theta)(1-\lambda) \geq 0$, so the sum is maximised termwise by $\text{own}(z) \in \arg\max_i S_i(z)$ over the candidates. □

Read plainly: **the objective is "balance the territories", plus "where there is slack, leave business with the representative who already has it".** That is a reassuringly operational reading of a Nash bargaining criterion, and it is exact.

5.1 Sizing the two terms

The relative weight of the two halves of (8) decides how much the legacy books can move the map at all, and it is set by *saturation* $t_z = T_z/M_z$. At the reference parameters $\theta = 0.40$, $\lambda = 0.30$ —

so $c_1 = 0.70$, $c_2 = 0.28$, $c_1 - c_2 = 0.42$ — and a saturation of ~5%, a zip whose whole book sits with one incumbent is worth

$$u_{\text{incumbent}} = 0.335\, M_z, \qquad u_{\text{other candidate}} = 0.314\, M_z.$$

Of the incumbent's utility, 89.6% is the pure opportunity term λM_z and only 6.3% is the incumbency premium $(c_1 - c_2)S_i(z)$; the utility swing between holding the book and not holding it is 6.7%.

The consequence for the algorithm is decisive. Combining Propositions 3 and 5 through the AM–GM inequality of (7),

$$\sum_i \log g_i \;=\; n \log\!\Big(\frac{W_0 + (c_1 - c_2)\sum_z S_{\mathrm{own}(z)}(z)}{n}\Big) \;-\; \underbrace{\Big[n \log \bar{g} - \sum_i \log g_i\Big]}_{D(g)\;\geq\;0}, \tag{9}$$

where $\bar{g}$ is the arithmetic mean of the gains and $D(g) \geq 0$, the log of the arithmetic-to-geometric mean ratio, vanishes exactly at perfect balance. The first term is bounded within a 6.3%-scale window by the incumbency premium; the second is unbounded and blows up as any territory is starved. Balance dominates, and incumbency is the tie-break — which is precisely the two-stage scheme of the next section, now derived rather than assumed.

This ratio must be checked against real saturation. At 5% the map is drawn by opportunity with a modest continuity tilt; at 30% it would not be.

6 The two-stage scheme

stage	problem	status
1 — draw	k balanced contiguous districts on opportunity alone	the hard part
2 — match	assign retained representatives to districts	built; exact

6.1 Stage 2 is a linear assignment problem on logs

Given a drawn map $A_1, \ldots, A_k$, let

$$g_{ij} \;=\; \sum_{z \in A_j} u_i(z) \tag{10}$$

be what district j is worth to representative i — evaluated for *every* representative on *every* district, unrestricted by legacy candidacy, which is the entire point of drawing the map before staffing it.

Proposition 6 (Nash-optimal staffing is a linear assignment problem)**.** *Suppose $g_{ij} > 0$ for all i, j. Maximising $\sum_i \log g_{i\sigma(i)}$ over injections σ from representatives to districts is a maximum-weight bipartite matching with weights $w_{ij} = \log g_{ij}$, and is therefore solved exactly in $O(\max(m,k)^3)$ by the Hungarian algorithm, where m is the number of representatives. When $m > k$ the matching is rectangular, and the unmatched representatives are exactly those not staffing the channel.*

Proof. The objective is additively separable over the pairs $(i, \sigma(i))$ that σ selects, the coefficient of (i, j) being $\log g_{ij}$, independent of the rest of σ. The injections σ are exactly the matchings of the complete bipartite graph $K_{m,k}$ saturating its smaller side. Hence this is the linear assignment problem with weight matrix $(\log g_{ij})$, which the Hungarian algorithm solves to optimality in cubic time; the rectangular case is the standard padding reduction. Positivity of g_{ij} is needed for $\log g_{ij}$ to be finite, and is checked rather than assumed. $\square$

Note that this is genuinely a different objective from utilitarian matching, and the difference is the reason to insist on it.

Example 1 (Nash and utilitarian matching disagree)**.** Two representatives, two districts, $g = \begin{pmatrix} 100 & 10 \\ 90 & 1 \end{pmatrix}$. The identity matching has utilitarian value 101 and Nash value 4.605; the swap has utilitarian value 100 and Nash value 6.802. The utilitarian rule picks the identity, handing representative 2 a district worth 1 to them because the *total* looks marginally better. The Nash rule picks the swap. A staffing that hands a senior wholesaler a territory containing almost none of their book is exactly the failure mode the channel cannot afford.

6.2 The cost of splitting, and the portfolio mitigation

Stage 1 cannot see relationships, so a good staffing may simply not be available on the map it draws. This is the same objection the programme raises to "decouple fairness from compactness": a two-stage scheme *relocates* the difficulty rather than removing it, and the joint optimum over (map, staffing) is in general strictly better than the sequential one — it is attained by first-stage maps that a balance-only stage 1 has no reason to prefer among its own optima.

The mitigation is cheap precisely because stage 2 is milliseconds: generate a *portfolio* of stage-1 draws — near-optimal balanced maps, which a branch-and-cut solver produces as a by-product of its incumbent sequence, or which a recombination-style sampler produces directly (DeFord et al., 2021) — score each by its best staffing value, and keep the best pair. This is not the joint optimum and should not be reported as one; it is a cheap lower bound on it, and the spread between the best and worst draw in the portfolio diagnoses how much the split is costing.

A pleasant side effect: stage 1 needs almost none of the confidential data — only (z, M_z) and the adjacency graph, with opportunity plausibly third-party market sizing — while stage 2 needs the books but not the geometry, and can therefore run entirely on the work machine given only the returned district map. The descaled export of Section 3 remains the right channel for anything that does need to travel, but this problem needs a fraction of it.

7 Disconnection and the balance ceiling

7.1 The bound

Let G have connected components $K_1, \ldots, K_C$ with masses $M_c = \sum_{z \in K_c} M_z > 0$. A contiguous district is connected, hence contained in a single component. So a partition of Z into k contiguous districts induces a *district budget* $(k_1, \ldots, k_C)$ with $k_c \geq 1$ and $\sum_c k_c = k$; feasibility requires $k \geq C$.

Proposition 7 (Balance ceiling)**.** *Every partition of Z into k contiguous districts satisfies*

$$\sum_{j=1}^{k} \log M_j \;\; \le \;\; \mathrm{UB}(k) \;:=\; \max_{k_c \ge 1,\; \sum_c k_c = k} \; \sum_{c=1}^{C} k_c \log \frac{M_c}{k_c}, \tag{11}$$

and $\mathrm{UB}(k) = -\infty$ *(no feasible partition exists) when* $k < C$*.*

Proof. Fix a contiguous k-partition and let (k_c) be the budget it induces; $k_c \ge 1$ because K_c is nonempty and must be covered, and $\sum_c k_c = k$ because each district lies in exactly one component. The districts inside K_c partition K_c, so by Proposition 3 applied within the component, $\sum_{j \subseteq K_c} \log M_j \le k_c \log(M_c/k_c)$. Summing over c bounds the partition's objective by $\sum_c k_c \log(M_c/k_c)$, which is at most the maximum over budgets. If $k < C$ then some component receives no district and is uncovered, so no partition exists. □

Two things make (11) valuable out of proportion to its difficulty. It is a *dual bound available before any solver runs* — it needs the component masses and nothing else, so four numbers suffice — and it is *tight in the right way*: it assumes a perfect split inside each component, so any shortfall it reports is geometric and no algorithm can recover it. The implementation (`channel.allocate_districts`) enumerates budgets directly, and `test_ceiling_is_an_upper_bound_on_any_real_partition` checks the inequality against brute force on every contiguous three-way cut of a path.

Corollary 8 (Small components force undersized districts)**.** *If some component has $M_c < \tau$ for a target territory size τ, then every contiguous partition contains a district of mass at most $M_c < \tau$. In particular a $\pm\epsilon$ band around τ is infeasible whenever $M_c < (1-\epsilon)\tau$, for every k and every algorithm.*

Proof. The districts inside K_c partition it, so the smallest has mass at most $M_c/k_c \le M_c$. □

Because the reported *spread* matters commercially as much as the objective, one more statement is worth making precisely, since it is weaker than the objective bound.

Proposition 9 (Spread floor)**.** *Fix a district budget (k_c). Every contiguous partition realising it has*

$$\frac{\max_j M_j - \min_j M_j}{M(Z)/k} \;\ge\; \frac{\max_c M_c/k_c - \min_c M_c/k_c}{M(Z)/k},$$

the right-hand side being the spread of the ceiling for that budget. Minimising over budgets gives an unconditional lower bound on the spread of any contiguous k-partition.

Proof. Inside component c the k_c district masses average M_c/k_c, so the largest is $\ge M_c/k_c$ and the smallest is $\le M_c/k_c$. Taking the maximum of the former over c and the minimum of the latter over c gives $\max_j M_j \ge \max_c M_c/k_c$ and $\min_j M_j \le \min_c M_c/k_c$; the mean $M(Z)/k$ is fixed. The unconditional statement follows by minimising over the finitely many budgets. □

Remark 2 (The two optima need not coincide)**.** The budget maximising the objective (11) is not always the budget minimising the ceiling spread, so the two bounds are reported from different budgets. In the table below this happens in exactly one cell: at $k = 6$ the east-heavy split's objective-optimal budget 1/3/1/1 has ceiling spread 87.1%, while the budget 2/2/1/1 has ceiling spread 77.4% at a lower objective. Both are valid bounds of their respective kinds; neither dominates.

7.2 What the ceiling says about this instance

The regional totals for the \$6.2B footprint are the outstanding input — four numbers, requiring no solver and no confidential per-zip data. Until they land, the table below is illustrative, using three plausible splits across the 4 components (west coast / east coast / Texas / Florida, in \$B). Each cell is $100\times$ the ceiling spread $(\max - \min)/\text{mean}$ over districts at the objective-optimal budget, computed by `channel.allocate_districts`.

split (\$B: W / E / TX / FL)	$k = 6$	$k = 7$	$k = 8$
even-ish (2/2.2/1.1/0.9)	**19.4%**	41.4%	55.9%
east-heavy (1.6/3/0.9/0.7)	87.1%	33.9%	**25.8%**
coast-heavy (2.6/2.6/0.6/0.4)	87.1%	101.6%	**60.2%**

Three conclusions, and the second is the one that changes a decision.

1. **\$1B ±10% is probably not reachable.** Even the friendliest split tops out at 19.4% ceiling spread, because a ∼\$0.9B region receives exactly one district (Corollary 8) and cannot be subdivided. Either the band becomes a target with a reported spread, or k moves — there is no third option, and no solver will find one.

2. **The best k depends on regional composition, and moves in opposite directions.** Under even-ish the ceiling is best at $k = 6$ and degrades as k grows; under east-heavy and coast-heavy it is best at $k = 8$ and *worst* near $k = 6$–7. So k is a *balance* decision informed by geography, not a headcount obtained by dividing \$6.2B by \$1B.

3. **This is four numbers of work.** Regional opportunity totals answer both questions immediately, with no solver and no confidential per-zip data.

8 Stage-1 formulation

Stage 1 is a balanced contiguous districting problem, the classical shape in the sales- and political-districting literatures (Hess and Samuels, 1971; Zoltners and Sinha, 2005; Ríos-Mercado and Fernández, 2009; Duque et al., 2011). Written in the n-way form the harness already carries, with $y_{z,i} = 1$ iff zip z is assigned to candidate i:

$$
\begin{aligned}
\max \quad & \sum_i \zeta_i \;-\; \rho \sum_{e\in E} w_e \\
\text{s.t.} \quad & \sum_{i\in \mathrm{cand}(z)} y_{z,i} = 1 && \forall z \in Z, \\
& g_i \;\le\; \sum_z u_i(z)\, y_{z,i} && \forall i, \\
& \zeta_i \;\le\; \log \hat g_i + \frac{\left(\sum_z u_i(z) y_{z,i}\right) - \hat g_i}{\hat g_i} && \forall i,\ \forall \text{ tangent points } \hat g_i > 0, \\
& \sum_{w\in C} y_{w,i} \;\ge\; y_{u,i} + y_{v,i} - 1 && \forall i,\ \forall u, v,\ C \text{ a } u,v\text{-separator}, \\
& y_{z,i} \in \{0,1\}.
\end{aligned}
\tag{12}
$$

Four notes, each of which is a trap already paid for in the two-player programme.

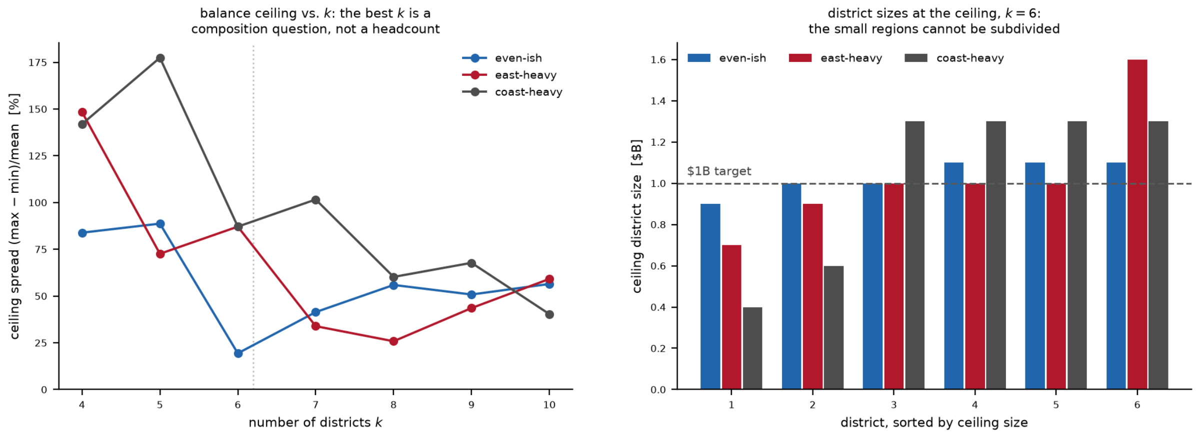

Figure 1: The balance ceiling of Proposition 7 on three illustrative splits of the \$6.2B footprint across its 4 components. *Left:* ceiling spread against k; the dotted line marks the k implied by a \$1B target. The curves cross repeatedly — there is no composition-free best k. *Right:* the 6 ceiling district sizes under each split, sorted, against the \$1B target. Under coast-heavy two districts are pinned below \$0.6B by components too small to divide, while the coastal districts sit at \$1.30B; no contiguous draw can improve on this picture, only fall short of it.

- **$\leq$, not $=$, on the gains.** The objective is increasing in g_i through the logs, so $g_i \leq \sum_z u_i(z) y_{z,i}$ is tight at every optimum. Writing it as an equality lets presolve multi-aggregate the gain variables out of the transformed problem, after which no in-callback incumbent can be posted at all. Relatedly, dual presolve reductions must be disabled: they count variable locks over the constraints presolve can *see*, and lazily separated rows are not there yet.

- **The tangents are outer approximation, one family per representative.** Each supports $\zeta_i \leq \log g_i$ at $\hat{g}_i$ and is valid everywhere, so the tangent set is a relaxation and the master's value is a genuine upper bound (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994); a certificate is the moment the incumbent meets it. The ε-certified piecewise-linear alternative (Vielma and Nemhauser, 2011; Caragiannis et al., 2019) trades the loop for a bounded a-priori error. The gain lower bound must come from the incumbent, never from an absolute floor like 10^{-9}: the log's gradient there is 10^9 and destabilises the LP.

- **The separator cuts are root-free and uniform.** If a feasible allocation puts both u and v with representative i, it contains an i-path between them; every such path meets the separator C, so some $w \in C$ has $y_{w,i} = 1$. If either endpoint is elsewhere the right-hand side is ≤ 0 and the cut is slack. Validity does not depend on how u, v, C were chosen, which is what lets the same construction separate fractional points. Crucially, *no root is fixed*: a rooted formulation is unsound on a disconnected graph, since it forces growth toward a vertex other components cannot reach, and its "dual bound" can sit strictly below feasible allocations already visited — not a corner case on a footprint that is disconnected by design. Encoding alternatives — single-commodity flow from a root (Shirabe, 2005), lazily separated cuts (Validi et al., 2021; Validi and Buchanan, 2022), cutting-plane aggregation (Oehrlein and Haunert, 2017) — differ mainly in how they pay for this.

- **Both cut families belong in *one* branch-and-cut tree.** Outer approximation converges by the Duran–Grossmann argument and lazy feasibility cuts by combinatorial Benders (Hooker and Ottosson, 2003; Codato and Fischetti, 2006); in a multi-tree loop that restarts a fresh MILP after each round, the two iteration counts *multiply.* A single tree with both families separated lazily at nodes (Quesada and Grossmann, 1992; Bonami et al., 2008; Validi et al., 2021) keeps one globally valid dual bound throughout.

The symmetry hazard, and the Hess remedy. Formulation (12) inherited its index i from the merger problem, where representatives are named and distinguishable. In stage 1 they are not: the k districts are *anonymous*, so any permutation of the labels maps a feasible solution to an equally good one and branch-and-bound must re-derive and re-prune $k!$ copies of every subtree — 720 copies at $k = 6$. The standard remedy is the centre-based formulation of Hess and Samuels (1971): introduce k *centres* chosen from the units, index districts by their centre, and require $y_{z,c} \leq y_{c,c}$. Labels are then pinned by geography and the symmetry is broken by construction, which is why the districting literature uses it (Zoltners and Sinha, 2005; Salazar–Aguilar et al., 2011; Ríos-Mercado and Pérez, 2012). The centre choice is itself a decision and interacts with contiguity, so this buys pruning with a larger model, not for free.

If 2,232 **units proves too large.** In increasing order of concession: per-component solves at $\sim$500 zips, which the disconnection already licenses; pre-aggregation to clusters or county/CBSA level, then disaggregation (Karypis and Kumar, 1998); ε-certified acceptance at the harness's existing tier-2 tolerance; and last a heuristic with a reported bound, for which Proposition 7 supplies the dual side for free and column generation or recombination sampling the primal (Gurnee and Shmoys, 2021; DeFord et al., 2021; Bozkaya et al., 2003). Connected fair division is NP-hard already for two agents with identical valuations (Deligkas et al., 2021), so exactness with certificates — not a closed form — is the realistic target; see Bouveret et al. (2017); Bilò et al. (2022); Suksompong (2019) for the fair-division-on-graphs setting and Bei et al. (2022) for the only price-of-connectivity bounds available (none is known for Nash welfare).

9 Open questions

These are modelling decisions, not implementation gaps. Each changes the answer.

1. **Empty bundles.** Insisting that every representative receive a non-empty contiguous bundle is a real and possibly infeasible constraint, and a single $g_i = 0$ sends $\sum_i \log g_i$ to $-\infty$. The standard treatment is lexicographic — maximise the count of representatives with positive utility, then maximise Nash welfare among them (Caragiannis et al., 2019) — and is the recommendation.

2. **Directionality of θ.** If capture depends on *which* representative is displaced, the single scalar becomes $\theta_{i\leftarrow j}$ and the identification problem grows from one parameter to $O(n^2)$ — and θ is already the one parameter resting on unfinished causal work.

3. **The brute-force acceptance tier.** The harness accepts a method partly on matching exhaustive enumeration at $n \leq 20$; at three candidates per zip that is $3^{20} \approx 3.5 \times 10^9$ leaves. The tier must be re-cut by candidate-weighted size $\prod_z |\text{cand}(z)| \leq 10^6$–$10^7$. This weakens a stated acceptance criterion and needs sign-off, not a quiet redefinition.

4. **Who owns an untapped or vacant zip?** Under $\mathrm{cand}(z) = \{i : S_i(z) > 0\}$ neither class has a candidate, yet both carry opportunity and both hold the graph together (the "zero-value glue" of failure regime (d)), so dropping them changes connectivity. Options: leave unallocated, assign by adjacency, or widen candidacy for these zips only. *For the national channel the question largely dissolves*: stage 1 partitions the whole footprint on opportunity alone, so every zip lands somewhere by construction, and candidacy re-enters only at stage 2 through (10).

5. **The filler capture mode.** $c_{\text{free}} \in \{c_2, c_1, \lambda\}$ per Section 2.3; `full` (c_1) is the recommendation. The three modes give materially different maps, so this is a decision and not a default.

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